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Research article

Evaluating environmental impacts of micro, mini and small hydropower plants in Norway

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ABSTRACT

Small-scale hydropower plants with an installed capacity <10 MW are an important part of Norway's local and regional development. Small-scale facilities have been viewed as a relatively environmentally friendly form of energy production because they are assumed to have limited negative environmental impact. However, the plants potentially have environmental impacts related to land use changes from infrastructure installation and also instream effects such as barriers to fish migration or disturbed flow conditions within bypasses. We used a remote sensing and GIS approach to examine the relative environmental impacts of three different size classes of Norwegian small-scale plants – categorized as **micro** (<0.1 MW), **mini** (0.1–1 MW) and **small** (1–10 MW). We evaluated 148 small-scale hydropower projects in Trøndelag, central Norway, focusing on their footprint and potential environmental impacts. Our analysis showed a slight preference for siting in forested habitat and, in particular, forested gorges associated with high biodiversity. However, most hydropower plants were not sited in forested gorges, or in areas with high predicted intensities of threatened terrestrial species. Similarly, there was little overlap with known migratory paths of Atlantic salmon and sea trout, or with archived occurrence records of instream fishes: salmon, trout or European eel. A low degree of overlap was attributed to the limited number of projects compared to the extensive distribution of these fish populations. The footprint areas of the projects were small (<4 ha) but increased with the size of the hydropower plants. Individual micro plants likely had low impact within their footprint because they required the least infrastructure for installation, and lacked dams which may act as barriers to fish movements. However, small plants were more land-use efficient. Our findings suggest that all size classes of small-scale hydropower plant have a role in balancing Norway's goals of minimizing environmental impacts of energy production while contributing to local and regional development.

1. Introduction

A societal transition to carbon-neutral renewable energies is central to energy policy, but future planning requires a better understanding of potential impacts on land use and natural resources (Dale et al., 2011; Dorber et al., 2019; IPBES et al., 2019; Rehbein et al., 2020). The topography and climate of Norway are optimal for hydropower production due to steep mountainous terrain combined with high annual runoff. Hydropower is the dominant energy source in Norway, with an installed capacity of 34 GW and an annual production of 138 TWh in 2022, accounting for approximately 88% of the total national production of 157 TWh (OED, 2022). Norwegian hydropower plants have diverse characteristics, ranging from small-scale, run-of-river HPPs, designed predominantly to produce power for local communities, to large-scale HPPs with inputs from impounded reservoirs or pumped storage. Most large-scale hydropower projects (≥10 MW) were developed during an early expansion period between 1950 and 1985 (NVE, 2008, NVE, 2017), whereas recent developments have mainly been small-scale hydropower plants (<10 MW) (Bakken et al., 2014; Lia et al., 2015). Energy consumption in Norway is forecasted to increase to 190–300 TWh/year by 2050 (Statnett, 2023), and new hydropower developments could help to meet future demand. A few undeveloped sites outside of protected areas could be developed for large-scale hydropower, or existing facilities could be upgraded with refurbishment of turbines or new extensions (Vilberg et al., 2023). Watersheds that are currently protected areas have been proposed for hydropower

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development (OED, 2015), but new large-scale projects are controversial and have not been adopted by the national government (Norwegian Government, 2022). Conversely, ca. 9500 potential sites have been identified as suitable for small-scale hydropower projects by the Norwegian Water Resources and Energy Directorate (hereafter 'NVE'), so small-scale hydropower will likely continue to play a role in Norway (Jensen et al., 2004).

Small-scale hydropower plants in Norway are classified into three size classes by NVE based on their installed capacity for power production. The smallest size class, *micro plants* (<0.1 MW), are typically sited along a river course and usually receive and discharge water from a diversion located near the plant. Micro plants are often owned and managed by farmers and local businesses who share energy produced between local requirements and the national grid. Connected to the grid through the owner's property, the owner can sell surplus power to the municipal power company when producing more electricity than required in summer, or buy power when local production is in deficit during low river flows in winter. Micro plants require the least expense for construction (not requiring burying the pipeline) and the widest range of solutions for connection to the grid, and installation costs may be reimbursed in less than a decade. The larger *mini plants* (0.1–1 MW) and *small plants* (>1 to <10 MW) almost always include an intake pipeline or 'penstock' draining an upstream location and may also have a dam upstream of the intake. Dam structures usually direct water to the intake rather than acting as storage reservoirs. An intake pond may also be present to prevent ice or debris from blocking intakes (Lia et al., 2015). Mini and small plants are designed for use with high heads (≥30 m) and low water flows (<1 m^3/s) so typically have one Pelton impulse turbine or two Francis reaction turbines for efficient power generation (Zidonis et al., 2015). Mini plants are operated by either local businesses for local energy demands or by hydropower companies, whereas almost all small plants are operated by hydropower companies. Flow modification in mini and small plants may be more pronounced than in micro plants because about a quarter of mini and small plants are licensed for hydropeaking. Licensing requirements for small-scale hydropower plants are less restrictive than for large-scale projects. Large projects (≥10 MW) require national government approval, while small projects (<10 MW) are approved by NVE, and micro and mini projects (<1 MW) are under local municipal authority (Thaulow et al., 2016). Thus, simplified regulatory procedures favour development of small-scale hydropower within Norway, with less "push-back" from local communities. Small-scale plants are often located in rural areas, and may have higher levels of social acceptance because local landowners receive financial and social benefits from the projects (Rygg et al., 2021). Indeed, a key aspect to the development of small-scale hydropower in Norway is to contribute to local and regional development (OED, 2007; Jonnasen, 2017).

An individual small-scale hydropower plant will have much less environmental impact than a large-scale plant, but past studies in Norway and other countries have found that multiple small-scale plants can potentially have greater cumulative effects compared to a single large-scale plant (Bakken et al., 2012; Kibler and Tullos, 2013; Koutsoyiannis, 2011). Conversely, development of a large-scale plant with a network of sites in an already-developed basin could result in limited damage whereas new construction of small-scale plants in undeveloped catchments could result in major environmental impacts. For instance, the potential for conflict with red-listed species may be greater for small-scale than large-scale plants due to the local requirements of where small-scale plants need to be sited (Bakken et al., 2014).

A knowledge gap for planning of renewable energy development in Norway is the lack of information on the comparative environmental impacts of different classes of hydropower plants (Bakken et al., 2012). Small-scale projects can cause various adverse environmental impacts, including disturbance of terrestrial biota from construction, and disruption of instream biota from changes in river flow, habitat and connectivity (Abbasi and Abbasi, 2011; Anderson et al., 2015; Kuriqi et al., 2021). Installation of plant infrastructure on land usually involves forest clearing for construction of penstocks and access roads, which causes disturbance of terrestrial biota (Bakken et al., 2014; Lia et al., 2015). Penstocks are often buried for landscape and aesthetic reasons, but construction can lead to encroachments that differ from the original cover (Erikstad et al., 2011). For example, areas disturbed by construction may have regrown herbaceous or shrubby vegetation rather than tree cover (Fig. 1). Small-scale hydropower plants are often constructed in forested river gorges, which are sheltered environments that often support high biodiversity (Evju et al., 2011; Olsen et al., 2020). Indeed, forested gorges have been classified as Valued Environmental Components (VECs) and were previously ranked as a near-threatened (NT) habitat on the 2011 Norwegian Red List for Ecosystems (Erikstad et al., 2020; Lindgaard and Henriksen, 2011). Finally, instream effects within a watercourse can include direct effects associated with building the plant, as well as prolonged effects such as dam obstructions to fish movements (Rivinoja et al., 2010), reduced fish habitat availability (Bakken et al., 2023; VRL, 2022), and fish mortality from flow regime changes (L'Abee-Lund and Otero, 2018).

Here, we evaluated some of the potential environmental effects for 148 small-scale hydropower plants in the county of Trøndelag in mid-Norway. Previous studies have considered small-scale plants in the nearby counties of Vestland (Bakken et al., 2012) and Nordland (Erikstad et al., 2020), and across Norway (Helland et al., 2011), but have not evaluated the relative impacts of different size classes of small-scale hydropower plants. Our project objective was to evaluate the relative overall environmental impacts of three different classes of small-scale hydropower plant (micro, mini and small). Hydropower can have multiple environmental effects, but we focused on three key types of natural resources that are most relevant to Norwegian conservation strategies and therefore best documented. First, we assessed the area of potential impact (the *footprint area*), and how the footprint was related to energy production for small-scale hydropower facilities. Second, we

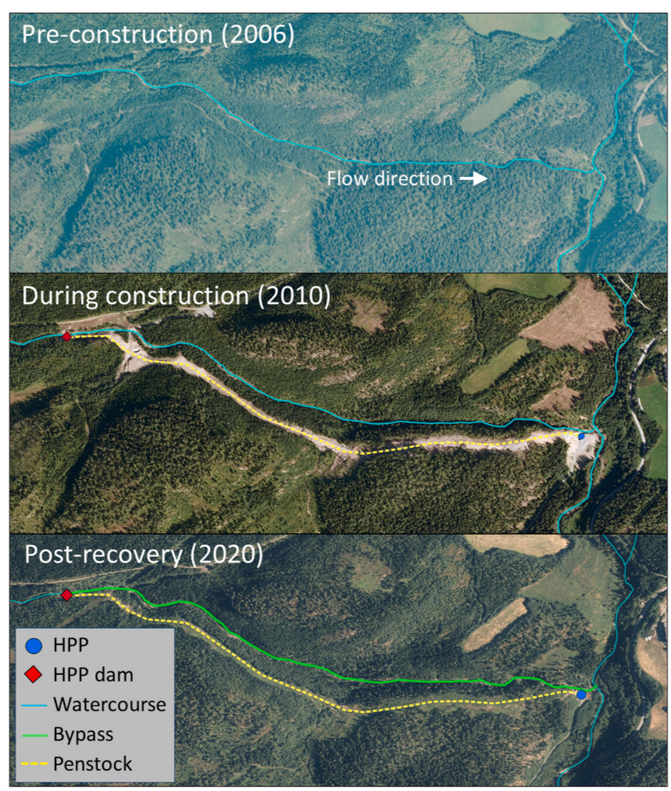

Fig. 1. Example of infrastructure for a small hydropower plant in Trøndelag (HPP Tua, Nr 1451) overlaid on images, pre-construction, during construction and post-recovery. Source: NVE.

assessed the potential ecological impact within the footprint area with respect to high biodiversity habitats including forests and forested gorges, and also occurrence records for terrestrial species of conservation concern. Last, we evaluated potential impacts on instream fishes using mapped reaches of anadromous salmonids (Atlantic salmon, *Salmo salar*, and sea trout, *S. trutta*), and archived observation records of brown trout, *S. trutta* (both the anadromous sea trout form and the river resident form), and European eel, *Anguilla anguilla*. Our new spatial analyses enabled us to assess environmental impacts of three size-classes of hydropower plant within the context of national plans for balancing sustainable energy production with needs for regional development.

2. Data sources

Quantification of the environmental impacts of the hydropower plants required an evaluation of the project infrastructure within the context of environmental properties. Information on the current state of small-scale hydropower plant infrastructure across the study region (Trøndelag, Norway) was compiled from an **NVE hydropower dataset** ("Vannkraft, Utbygd og ikke utbygd", https://nedlasting.nve.no/gis/) (compiled on Feb-2022). The NVE dataset included detailed information on locations and characteristics of: (1) hydropower plants (Fig. 2); (2) hydropower dams; (3) hydropower pipelines including penstocks; and (4) penstock intakes. The NVE dataset is continually updated for all operational and planned hydropower plants in Norway, and includes information on hydropower plant location, capacity, size class, and operational status. Hydropower plants can be cross-referenced with connected dams, penstocks and intakes using a unique hydropower ID. Estimates of annual energy production (GWh/year) for each plant were obtained from the NVE's updated production status reports.

(https://www.nve.no/energi/energisystem/vannkraft/status-for-ny-vannkraftproduksjon/). Information on the **river network** on which the hydropower plants were situated was also obtained from NVE (Elvenettverk/ELVIS). The river network contains the paths of most streams and rivers in Norway, and identifies sections of individual streams using a Strahler stream order (1 = upstream, lower order tributary, 2 = downstream, higher order tributary, etc.).

The dominant **landcover** at the site of the hydropower plants was obtained from a 30 m resolution satellite imagery-derived raster vegetation map (satveg, Johansen, 2009). This is the finest resolution comprehensive and continuous landcover raster map available for Norway. Estimating landcover at higher resolutions, such as 10 m, was

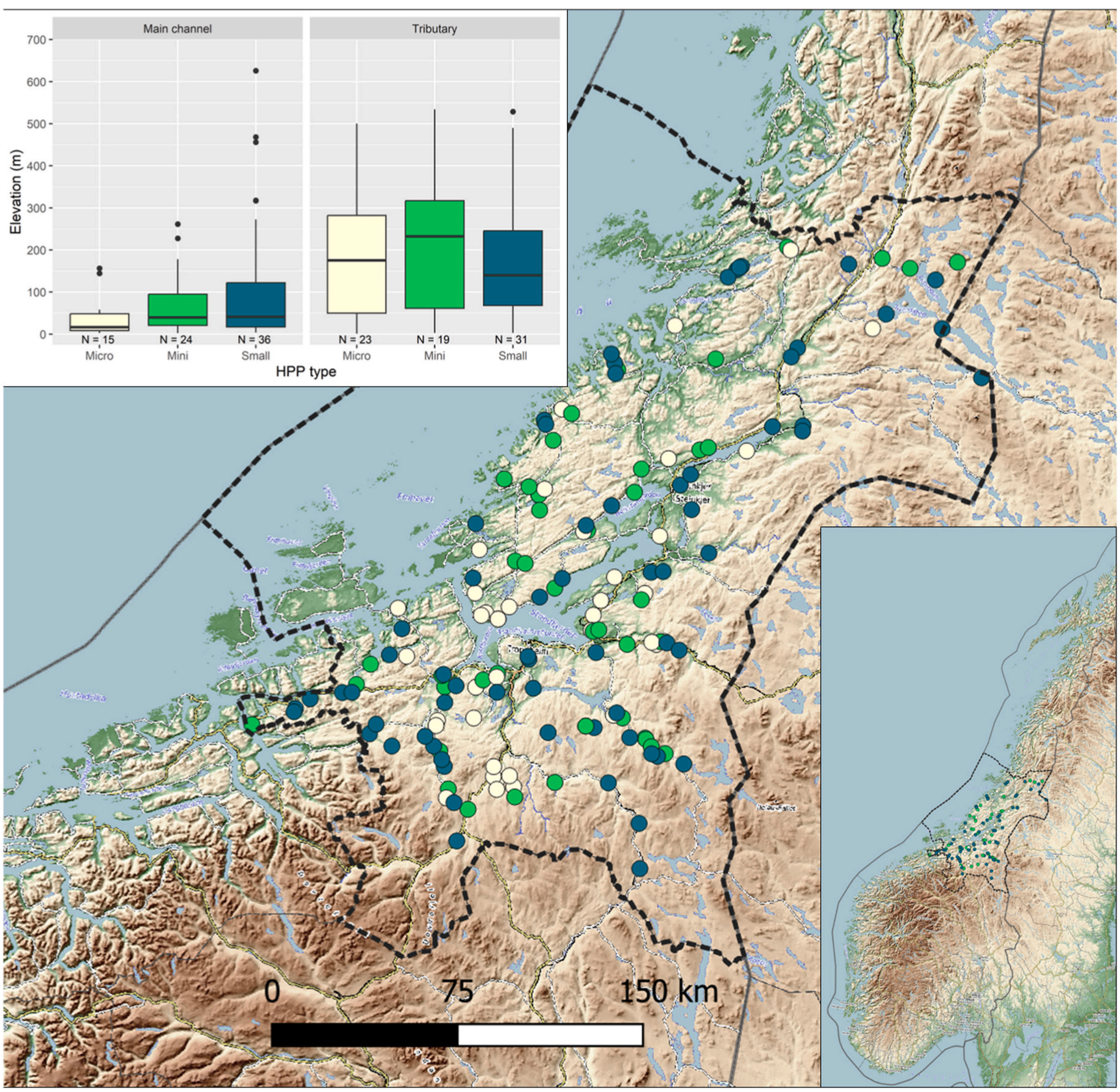

Fig. 2. Locations of the 148 small-scale hydropower plants (HPPs) in Trøndelag, central Norwegian Government, 2022. Trøndelag county boundaries are shown by the black dashed line. The background map is OpenTopoMap (opentopomap.org).

deemed unnecessary for the broad landcover categories used in this study. The vegetation map classifies landcover into 25 different types, but for our purposes, we combined different landcover categories into eight broad classes: forest, heather/meadow, heathland, bog/marsh, ice/snow/exposed, cultivated, urban, and other. **Surface topography** was obtained from a 10 m resolution airborne topographic LiDAR-derived digital elevation model (DEM), available through a data portal of the Norwegian Mapping Authority (https://hoydedata.no/LaserInnsyn2/). This resolution was chosen as it provides a balance between detail and computational efficiency. Changes in landcover from pre-to -post hydropower plant construction were derived from **aerial and satellite imagery**, obtained from a data portal of the Norwegian Mapping Authority (Norge i bilder, https://norgeibilder.no/). Imagery was available for most sites on a yearly basis (or more frequently), enabling year-to-year variation to be determined. The spatial distributions of **threatened terrestrial species** were obtained from maps of conditional intensity derived from point process models for five taxonomic groups of organisms: mosses, lichens, vascular plants, fungi, and arthropods (arachnids and insects) (compiled by Olsen et al., 2018, 2020). The taxa included species that were listed as being of conservation concern (critically threatened, endangered and vulnerable) on the Norwegian Red List for Species.

We compiled two data sources to assess potential impacts of small-scale hydropower on stream populations of fishes: maps of reaches for anadromous salmonids, and archived observation records of fish occurrence. First, detailed maps of the distribution of **anadromous salmonid reaches** were obtained from recent national surveys in Norway for anadromous populations of Atlantic salmon (N = 89 reaches) and sea trout (N = 138 reaches) (Fiske et al., 2024; Forseth et al., 2019; VRL, 2022) (see https://lakseregisteret.fylkesmannen.no/). Second, archived records of fish occurrence were obtained from Artsdatabanken (https://www.artsdatabanken.no/), a Norwegian portal for natural history records. We compiled locality records of brown trout, which includes both the migratory and resident forms, and locality records of Atlantic salmon and European eel. Occurrence records were contributed by five research organizations engaged in field sampling including the Norwegian Institute for Nature Research, the UiO Natural History Museum, the Environment Agency, the NTNU Science Museum, and the Norwegian Zoological Association. Observations have been registered since 1870, but 99% been made since 1960, and 22% since 2000.

3. Assessing environmental impacts of small-scale hydropower

Environmental impacts of small-scale hydropower plants were assessed using a remote sensing/GIS approach based on QGIS and the *terra* library of R (https://rspatial.org/index.html). First, *footprint area* was calculated for each plant. The footprint included the dimensions of impacted areas related to the construction extent on land and the length of the instream bypasses. The estimates provided an approximate metric of area of impact, with the caveat that some land-based construction of roads and other infrastructure might be for multi-use purposes (Erikstad et al., 2011), whereas some effects on watercourses may extend downstream of bypasses. Footprint area was compared with annual energy production – *footprint area* per *unit energy production* – to determine which class of hydropower plant (micro, mini, small) had the most efficient land use and thereby minimized footprint area for a given level of energy production. Second, the potential environmental impacts within the footprints of the plants were assessed with respect to overlap with major terrestrial and instream environments. Terrestrial impacts were assessed from: (1) the overlap between hydropower plants and forest habitats as well as forested gorges that are known to be important in terms of high biodiversity, and (2) the overlap between hydropower plants and areas of high intensity for occurrence of threatened terrestrial species. Next, (3) the instream impacts were assessed by quantifying overlap between hydropower plants and salmonid reaches mapped for migratory populations of Atlantic salmon and sea trout, and fish distributions estimated from archived recordings of Atlantic salmon, migratory and resident brown trout, and European eel. We focused on these three key fish species because of their high socio-cultural importance within Norway (Liu et al., 2019; Myrvold et al., 2019).

3.1. Footprint area

Land footprint. The land footprint area from the installation of mini and small hydropower plants was calculated using a similar approach to Bakken et al. (2014). For each site, we examined remote sensing imagery over a time-period that spanned construction of the plant. We selected images with an acquisition time as close as possible to the timing of plant construction, with images from before and after development typically <5 years. We also prioritized use of high resolution aerial photos (10–20 cm resolution) over Sentinel-2 images (10 m resolution) because aerial photos showed more detail, but Sentinel-2 images were occasionally used if aerial photo coverage was not available. We then manually digitized areas with observed landcover change from the pre-to post--installation periods for each plant using QGIS. Landcover changes used to assess the land footprint included construction of turbine housing, infrastructure, access roads, clear-felling of forest swathes for installation of pipelines, and deposits of rock debris produced by boring of tunnels. The land footprint for each mini and small plant was then calculated as the manually digitized polygon area. It was generally not possible to observe land footprints for micro plants within the imagery. We therefore relied on information on pipeline length registered in the NVE hydropower dataset to make direct comparisons of efficiency among all three size classes of small-scale hydropower plant.

Instream footprint. The instream footprint from the bypass extent was the reach where flows are likely to be most affected by extraction of water for the hydropower plant. The bypass was measured as the length of the river network (source: Elvenettverk/ELVIS) from the location of the plant intake to the downstream end of the bypass (source: NVE hydropower dataset). For cases where both the intake and outlet were located on the same reach, the downstream end of the bypass was determined from the intersection of the outlet pipeline within the river or from the location of the plant. For rivers where the plant outlet was situated in a fjord, reservoir, or a downstream location in a higher order river, the bypass length was measured as the distance from the intake location to the mouth of the tributary containing the plant.

Differences in land and instream footprints among the hydropower size classes (micro, mini and small) were assessed using nonparametric Wilcoxon tests (R-function *wilcox_test{rstatix}*) and Kruskal-Wallis tests (*kruskal_test{rstatix}*). Nonparametric tests were used due to the non-normal distribution of footprint data (confirmed through Shapiro-Wilk tests for normality; R-function *shapiro.test{stats}*) and inhomogeneity of variances in footprint area among size classes of hydropower plant (confirmed using Levene's test; R-function *leveneTest{car}*). After applying Kruskal-Wallis tests, Wilcoxon tests using a Bonferroni adjustment were then used to make pairwise comparisons among hydropower classes. The same approach was also used to assess footprint area as a function of the annual energy production.

3.2. Potential environmental impacts

Impacts on terrestrial biota. The potential impacts of small-scale hydropower plants on terrestrial biota within the footprint area were determined by: (1) assessing the overlap between hydropower plants and two landcover types associated with high biodiversity (forest habitat, and forested gorges), and (2) the overlap between hydropower plants and the distributions of threatened terrestrial species. Landcover data, providing information on forested habitat, were obtained from the 30 m resolution *satveg* vegetation map. The occurrence of forested gorges was determined from the overlap between forested habitat and gorges. Gorges were identified from local depressions in the 10 m resolution DEM, using an approach based on Erikstad et al. (2020). All pixel locations within the DEM were defined as being a gorge when: (1) the average elevation of the pixel was ≥10 m less than that of the mean of all surrounding pixels within a window (110 × 110 m) centred on the pixel; and (2) the distance to the nearest river location was <100 m. The second criterion was included to ensure that gorges were only identified near to rivers, and that landscape features were excluded if they were distant from river courses. The spatial distribution of threatened terrestrial species was determined from maps of conditional intensity of five taxonomic groups (Olsen et al., 2018, 2020). An overall intensity of occurrence for threatened species was then calculated as the sum of intensities across each of the five taxonomic groups.

We then examined whether hydropower plants were more likely to be sited in forest habitat, forested gorges or areas with a high intensity of threatened species than would be expected based on background levels. First, we determined the landcover and the intensity of threatened terrestrial species within the vicinities of the hydropower plants using a 100 m radius around each plant. For each plant vicinity, we determined the dominant landcover (forested versus non-forested habitat, and forested gorge versus "other") and the mean intensity of threatened terrestrial species. We then compared distributions in the hydropower plant vicinities with the respective background levels from the entire region of Trøndelag outside these 100 m radius vicinities using one-proportion Z-tests for forest habitat and forested gorges (R-function *prop.test{stats}*) and one-sample Wilcoxon signed range tests for species intensity (R-function *wilcox.test{stats}*). Differences among the three size classes of hydropower plants were then examined to determine if landcover and intensity of threatened species depended upon size class (R-functions *Prop_test{rstatix}* and *kruskal.test{stats}*, respectively).

Impacts on instream fishes. First, using data on mapped anadromous salmonid reaches, the NVE hydropower dataset, and manual investigation of remote sensing images, we determined the overlap between small-scale hydropower and anadromous salmonid reaches, both for salmon rivers and trout rivers. We determined the percentage of salmonid reaches where a hydropower plant overlapped with the reach. We then identified the percentage of reaches containing small-scale hydropower plant dams crossing the river channel, on the rationale that the presence of dams would create barriers to upstream movement of the anadromous fish. Second, using our measurements of bypass dimensions and species occurrence records from Artsdatabanken, we determined the overlap between small-scale hydropower bypasses and occurrence records for Atlantic salmon, brown trout, and European eel. We filtered the Artsdatabanken dataset to restrict our sample to records from rivers or lakes (excluding those from the sea) for the three selected species. We then determined the percentage of observations for each of these species that were within or near to hydropower bypasses, using a 500 m buffer around each bypass based on Helland et al. (2011). This allowed us to assess the overlap between the observed distributions of the three species across Trøndelag county and the locations where the bypasses were situated.

4. Results

A total of 148 small-scale hydropower plants (<10 MW) were registered in Trøndelag: 38 micro (<0.1 MW), 43 mini (≥0.1–1 MW) and 67 small (>1 to <10 MW). Together, the projects in Trøndelag constituted 10.3% of the 1434 small-scale hydropower plants registered in Norway. The oldest plant began operation in 1910 (the mini Skulbørstadfoss plant, *HPPnr* 1396), but nearly two-thirds of plants (N = 89, 60.1%) started operation after 2000. Plants were sited across Trøndelag (Fig. 2), in both the main channel (N = 75) and in tributaries (N = 73) of different water courses. Projects within tributaries were sited at significantly higher elevations (median = 175 masl) than facilities within the main channel (median = 38 masl) (Wilcoxon test, $p < 0.001$). Small-scale plants were mainly sited distantly from one another: only 30 plants were sited on the same watercourse as other small-scale plants; the median straight-line distance between neighbouring plants on the same watercourse was 4.6 km (range = 1.0–49.9 km). Assuming an average drop height and maximum intake per hydropower plant, the small-scale hydropower plants provided a total average energy production of 933 GWh/year or 9.9% of the total energy production from hydropower in Trøndelag (9438 GWh/year) in 2022, https://www.ssb.no/statbank/table/08308/).

4.1. Footprint area

The footprint area of small-scale hydropower plants was generally small, with an average construction area <4 ha, and average pipeline and bypass lengths <1.5 km (Fig. 3a–c). Footprint area increased with the size class of the plant, both on land and in instream habitats. Median construction areas for small plants were ca. 4.1 ha, more than twice that of mini plants at ca. 1.5 ha (Fig. 3a). Construction areas were not identifiable in the aerial and satellite imagery for micro plants, suggesting that the footprint areas would have been minimal in comparison to the larger mini and small plants. Median pipeline length increased with the size class of the hydropower plants from 410 m for micro plants, ca. 770 m for mini plants, to ca. 1100 m for small plants (Fig. 3b). Bypass length in the water course also increased with the size class of the plant, where the median lengths were ca. 470 m for micro plants, ca. 860 m for mini plants, and 1460 m for small plants (Fig. 3c). Footprint areas were therefore 2–3 times greater for the largest size class of small-scale hydropower plant than the smallest size class.

Land use efficiency measured as the footprint area per unit energy production was better for larger size classes of hydropower plants (Fig. 3d–f). Construction area per unit energy tended to be greater for mini plants than small plants, but was not significantly different (Fig. 3d). Pipeline and bypass lengths per unit energy were significantly smaller for mini plants than for micro plants, and significantly smaller for small plants than mini plants (Fig. 3e & f). Therefore, for the same amount of energy production, micro plants would have had a larger footprint area than mini plants, which in turn would have had a larger footprint area than small plants.

4.2. Potential environmental impacts

Impacts on terrestrial biota. Analyses of the landcover and the intensity of threatened terrestrial species within 100 m of the small-scale hydropower plants indicated that siting of hydropower plants tended to be in areas of higher biodiversity in the Trøndelag region (Fig. 4a–c). Circa 75% of hydropower plants were sited in forest habitat, which was significantly greater than the background value of ca. 45% forest cover within the region (Fig. 4a). Hydropower plants were also preferentially sited in forested gorges, with ca. 5% of hydropower plants being located in forested gorges, as opposed to a background prevalence of <1% within the Trøndelag region (Fig. 4b). Some hydropower plants had a higher intensity of occurrence for threatened terrestrial species than the background intensity across the region (Fig. 4c), but the mean intensity in the vicinities of the hydropower plants was not significantly greater than that of the region.

Impacts on terrestrial biota did not significantly vary according to size class of hydropower plants (Fig. 4d–f). Mini and small plants were more likely to be sited in forested habitats than micro plants (Fig. 4d) but the differences were not statistically significant. No significant differences existed among the three size classes of hydropower plants in their probability of being located in forested gorges (Fig. 4e) or in their overlap with sites with a high intensity of threatened terrestrial species (Fig. 4f).

Impacts on instream fishes. Anadromous salmonid reaches were largely free of small-scale hydropower plants. Of the mapped reaches (Atlantic salmon N = 89; sea trout N = 138), only 19.1% of salmon reaches and 8.7% of trout reaches had small-scale hydropower plants located within them. Hydropower plants of all size classes were found within the reaches, but small plants were most prevalent: within salmon

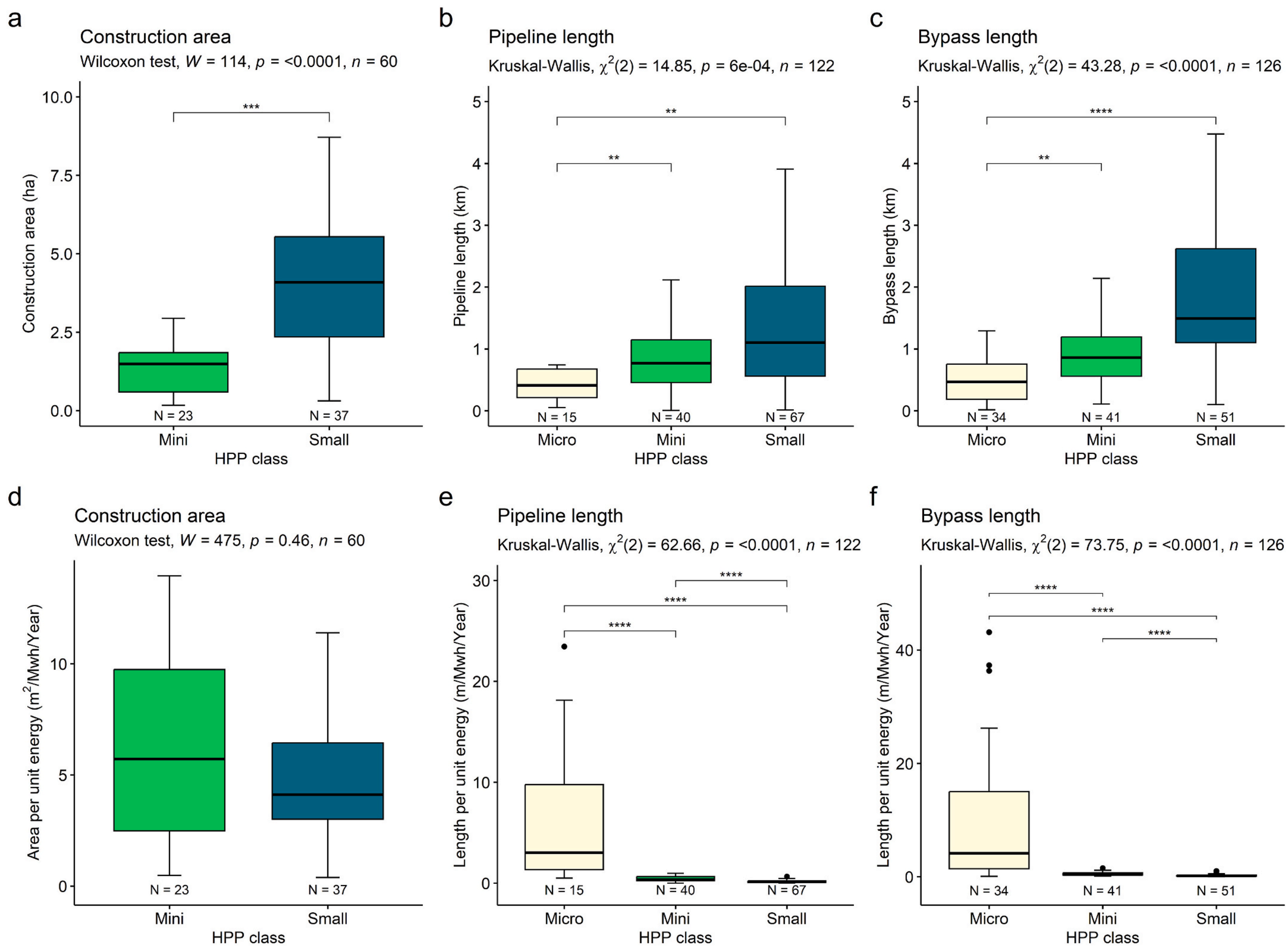

Fig. 3. Footprint areas of small-scale hydropower plants (HPPs) in Trøndelag, Norway: construction area, pipeline length and bypass length (a–c); and construction area, pipeline length and bypass length per unit energy production (d–e). NB: Wilcoxon test significance levels: p < 0.05 (*), p < 0.01 (**), p < 0.001 (***); non-significant pairwise comparisons are not shown.

reaches, N = 2 (micro), N = 3 (mini), N = 12 (small); within trout reaches, N = 4 (micro), N = 2 (mini), N = 6 (small).

Dams associated with small-scale plants, which could act as hard barriers to fish movements, were rare, only occurring in ca. 2% of salmonid reaches. Only three dams associated with small hydropower plants were found, all occurring at the top of salmonid reaches (1 in a trout reach, and 2 in reaches used by both salmon and trout). Dams were absent for micro and mini plants.

Overlap between the distribution of fishes from occurrence records in the Artsdatabanken database and bypasses for small-scale hydropower plants was minimal. Few locality records (<1–2%) of Atlantic salmon, brown trout and European eel were located within a 500 m buffer of the bypasses: 0.59% of Atlantic salmon observations (127 out of 21,570 records); 0.87% of brown trout observations (175 of 20,680); and 1.12% of European eel observations (6 of 535). Overlap between observations and bypasses was more frequent for micro plants (68.4% of observations that occurred within all the bypasses) than for mini plants (10.2%) or small plants (21.2%).

5. Discussion

Comparative information on the impacts of different forms of renewable energy is a knowledge gap that affects planning for meeting future needs for increased energy demand. Our spatial analyses for small-scale hydropower in Trøndelag county in Norway resulted in three new findings. First, the limited number of small-scale hydropower plants in Trøndelag minimized the aggregate overlap with critical conservation habitats, such as forested gorge habitats, or terrestrial and instream species of conservation concern. Second, small-scale hydropower plants had small footprint areas. Together, the two factors will have limited *aggregate* environmental impacts. Third, impacts within the footprint area were dependent on size class of the hydropower plants, although the gradient of the impacts depended on the environmental property. The total footprint area increased across the three size classes of small-scale hydropower (micro, mini, and small). Notably, micro plants (<0.1 MW) exhibited the highest footprint area per unit of energy produced, whereas small plants (1–10 MW) showed the least, indicating that the larger plants are more efficient in land use. Our findings highlight the limited but significant environmental impacts of small-scale hydropower in Trøndelag, that vary by plant size. We discuss these impacts and their implications for Norway's national energy goals, suggesting that while small-scale hydropower is land-efficient, its overall contribution to energy production remains limited.

5.1. Environmental impacts of small-scale hydropower in Trøndelag

Our spatial analysis using remote sensing and GIS data revealed that individual small-scale hydropower plants in Trøndelag typically have a

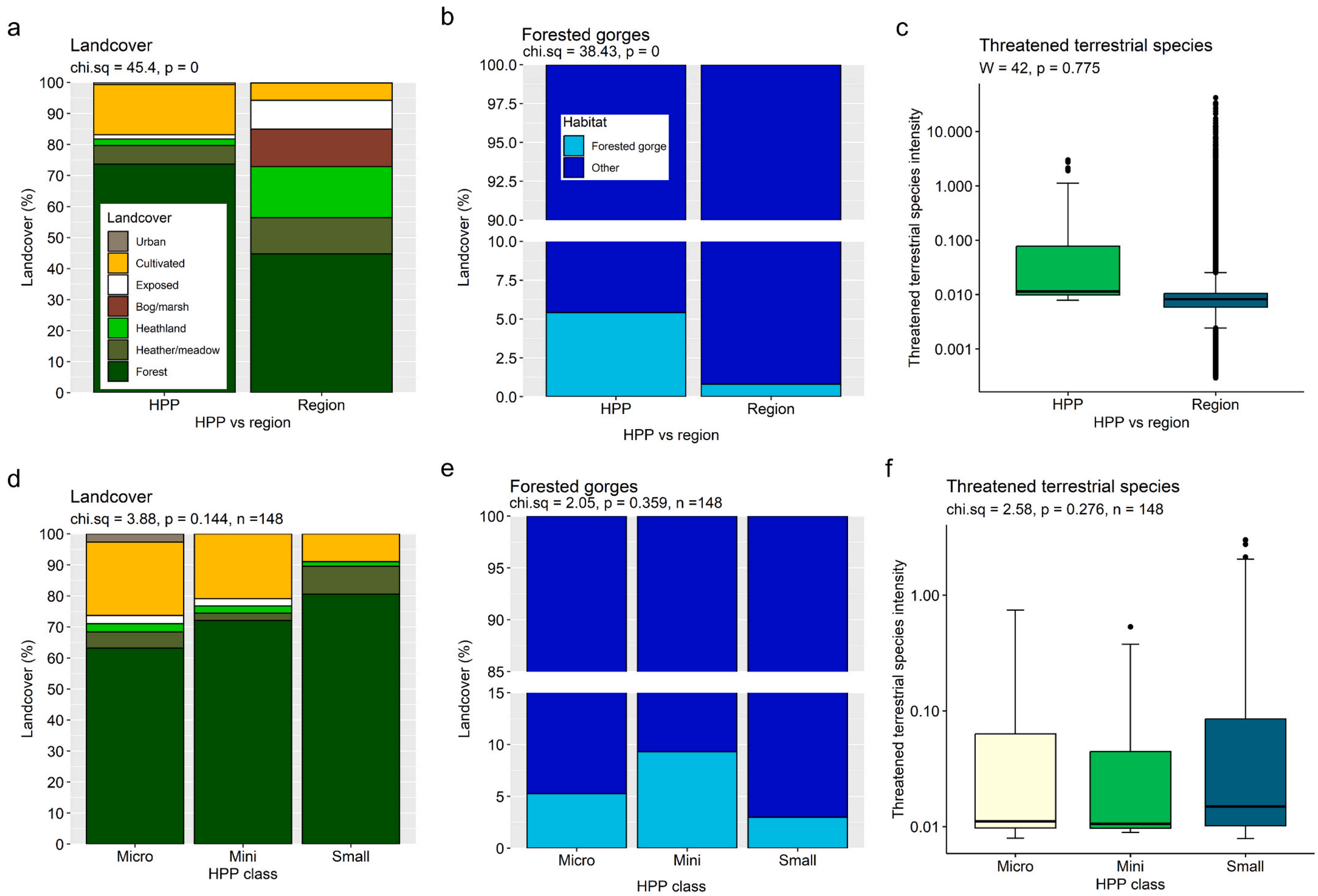

Fig. 4. Landcover, forested gorges, and intensity of threatened terrestrial species within a 100 m distance of small-scale hydropower plants (HPPs) in Trøndelag, Norway: all plants versus region for landcover type (a), forested gorges (b), and intensity of threatened terrestrial species (c); and plant size class for landcover type (d), forested gorges (e), and intensity of threatened terrestrial species (f). Note the y-axis break in panels (b) and (e), and the logarithmic scale in (d) and (d). Statistics presented in (a), (b) and (c) are for tests comparing whether the respective metric (forest proportion, forested gorge proportion, or mean intensity of threatened species) for the HPP is significantly greater than the background value for the region.

small footprint, averaging less than 4 ha in construction area, with pipeline and bypass lengths averaging under 1.5 km. Within these footprints, potential for environmental impacts on terrestrial biota and on instream fishes were present, but the total potential impact was limited by both the low number of hydropower plants and the relatively small individual footprint area per plant.

Impacts on terrestrial biota. Small-scale hydropower plants tended to be preferentially sited in forested locations, particularly forested gorges, and sometimes in areas with high intensity of occurrence for threatened species. Our findings are therefore consistent with previous studies from the nearby counties of Nordland (Erikstad et al., 2020) and Vestland (Bakken et al., 2014). Erikstad et al. showed that development of all potential sites for small-scale hydropower would impact a majority of forested gorge segments; whereas Bakken et al. found that that development of existing small-scale hydropower occurred in areas where threatened species have been observed. For the Trøndelag region, however, the current magnitude of ecological impacts was likely to be relatively low. First, about 95% of hydropower plants within Trøndelag were not sited in forested river gorges. Second, although some hydropower plants were sited in locations with high intensities of threatened terrestrial species, most were sited in locations with low intensities that were similar to much of the wider region. Thus, the *current* impact on the studied biota is small. However, development of hydropower inevitably involves loss or degradation of nature within the footprint area so could have impacts on other aspects of terrestrial biota not considered here.

Impacts on instream fishes. Small-scale hydropower plants had limited potential for impact on anadromous salmon or trout because a majority of salmonid reaches (salmon ca. 81%; trout ca. 93%) contained no small-scale hydropower plants. A lack of overlap coupled with the absence of dams for micro and mini plants, meant that only three dams associated with small plants could act as potential barriers to upstream movement. Few registrations of Atlantic salmon, brown trout and European eel occurred within locations where bypasses were constructed, suggesting that the wide spatial distribution of these populations across the region will mean that the species are not particularly impacted by flow conditions within the bypasses. Small-scale hydropower is generally considered to have high impacts on resident populations of riverine fish (Erikstad et al., 2011; Foldvik et al., 2019; Helland et al., 2011). However, the number of small-scale hydropower plants in Trøndelag is currently low, and most of the waterways in the county are unaffected by small-scale hydropower plants.

The environmental impacts identified, both on terrestrial biota and on instream fishes, provide an indication of some of the potential impact of small-scale hydropower within Trøndelag. Although our analyses utilized diverse data sources, they were constrained by incomplete coverage and potential biases inherent in the data quality. These limitations underscore the need for more comprehensive and detailed data on environmental conditions and biodiversity. Small-scale hydropower is known to have a wide range of environmental impacts, but data on many pertinent environmental properties are still lacking. For example, detailed maps of the instream populations of freshwater invertebrates do not yet exist for Trøndelag. Second, even when distribution data were available, the information was often incomplete. Records of the known occurrence of Atlantic salmon, brown trout, and European eel provided a broad-scale indication of the species distributions, but sampling coverage was incomplete so the potential overlap with small-scale hydropower could be greater than our estimates.

5.2. Relative environmental impacts of micro, mini and small hydropower plants

The overlap between hydropower plants and habitat and terrestrial biota that have been prioritized for conservation varied according to size class of the plants. Hydropower plants of the larger size classes tended to be sited more often in forested habitats – ca. 80% of small plants were in forested habitats, as opposed to only ca. 73% of mini plants and ca. 63% of micro plants. Footprint area increased with the size class of the plant, suggesting that when overlap occurred, hydropower plants of larger size classes will have greater impacts.

Overlap with instream fish populations also depended on size class. First, small plants were sited more often on salmonid reaches than micro or mini plants. Second, dams, which are potential migration barriers were only found for small plants. Conversely, registrations of fish presence within bypasses were more commonly associated with micro plants than mini or small plants. Thus, the effect of size class showed a mixed picture. However, the overall instream effect of micro plants is likely to be small because (1) they lacked dams so will have had less impedance on migration, and (2) are not designed for short-term energy balancing so will not have caused major flow disruption within the bypass from hydropeaking. Mini and small plants, with dams that can disrupt migration and hydropeaking that can modify flows, may have more of an impact. However, the small number of these plants limited their aggregate effects. Groups of plants could potentially have increased aggregate effect if areas of influence overlap and the plants are not managed together to take this into account. Regulations governing water flows within Norwegian mini and small plants are initiated on an individual basis, limiting the potential for integrated management of neighbouring plants, which are often owned by different power companies. However, most small-scale plants within Trøndelag are not co-located on the same watercourse. Even when hydropower facilities were located on the same watercourse, small-scale plants were separated by several kilometres, so plant bypasses were non-overlapping. This lack of overlap means that impacts can generally be considered on a plant-by-plant basis.

Variation among small-scale hydropower size classes existed in terms of both footprint area and land use efficiency measured as footprint area per unit energy produced. Of the three plant types, micro plants were least land use efficient as expressed via pipeline and bypass length per unit energy produced. However, this pattern does not necessarily imply that the development of micro hydropower plants is ill-advised, because the projects have a small aggregate impact, and the rationale behind their development could be justified based on economic needs or social policy. Aggregate impacts of micro plants will have been small due to both a small footprint area, and potentially less impact within this footprint. Micro plants typically had short pipelines and bypasses. The actual impact of the pipelines may have been limited given that they often only involved laying a PVC pipe on the ground surface rather than digging a trench for installing a more substantial pipeline. The impacts within the bypass may have been limited by the absence of a dam crossing the channel and a lack of hydropeaking. Thus, the small aggregate footprint, and possibly small impact within this footprint, may not necessarily outweigh the benefit of micro hydropower for decentralized energy production that contributes to local and regional development. For land-use efficient energy production that contributes to the national grid, however, the results of the current study would suggest prioritising small- over mini or micro plants. Our findings are crucial for informing policy and decision-making related to the implementation of small-scale hydropower within Norway's national energy strategy, highlighting the need for balancing energy production with environmental conservation.

5.3. Relevance of small-scale hydropower to Norway's national goals for energy production

The areal extent of small-scale hydropower is still relatively small compared to large-scale hydropower projects. The total construction area of mini and small plants in Trøndelag was 201 ha; by comparison, large-scale hydropower plants ($\geq$10 MW) within the same region had an average direct occupancy of 1065 ha per project (N = 15 projects, Kenawi et al., 2023). Despite low levels of implementation in Trøndelag, small-scale hydropower is land use efficient in comparison to large-scale hydropower projects. The mean footprint area per unit energy production for mini and small hydropower plants in Trøndelag was 4.7 m^2/MWh/year (range = 0.39–13.96, N = 60). Our estimates of efficiency are higher but comparable to findings of Bakken et al. (2014) for small-scale hydropower plants in Vestland (mean = 21.5 m^2/MWh/year, range = 3.4–96.1, N = 27). In comparison, alternative sources of renewable energy generation often have a much larger footprint per unit energy. For example, Bakken et al. (2014) reported footprints per unit energy for large hydropower (mean = 47.8 m^2/MWh/year, range = 36.6–62.7, N = 3) and land-based wind power (47.1 m^2/MWh/year, N = 1) that were an order of magnitude greater than our estimates for small-scale hydropower plants.

Despite small-scale hydropower being more land use efficient than large-scale hydropower, the potential role of small-scale hydropower in fulfilling Norway's energy needs is likely to be limited. Small scale hydropower currently provides ca. 8.6% (11.9 TWh/year) of Norway's hydropower production (138 TWh/year) and ca. 12% (1.0 TWh/year) of Trøndelag's hydropower production (8.3 TWh/year). Thus, small-scale hydropower contributes little total energy production in comparison to large-scale hydropower. Small-scale hydropower is also an intermittent source of energy so does not improve the stability of energy production in Norway. The promotion of small-scale hydropower, however, is ongoing because of its important role in local and regional development, allowing for decentralized siting, community involvement, and profitability for small farms (Rygg et al., 2021). The contributions are diverse (Jonnasen, 2017). Benefits include direct effects, such as dividends, taxes, fees, and direct employment, and indirect effects, such as investment costs for real estate companies and operating costs accruing to local suppliers. Both direct and indirect effects lead to further upshots for the local economy, such as additional employment in both public and private sectors. Effects will vary according to the hydropower plant in question, and will depend upon size class. A micro plant typically involves minimal construction so real estate companies might not be involved. Immediate economic benefits will likely be localized to the farmer or local business operating the plant, although the additional income source may benefit the local economy. A small plant involves more construction and real estate involvement, and may require local employment for plant and maintenance, so will have a greater impact on local and regional development. Thus, small-scale hydropower may have a minor role in fulfilling Norway's goals for energy production, but all small-scale size classes – micro, mini and small – will continue to play a role in local production of energy and contributing to local and regional development.

6. Conclusions

The total potential environmental impacts from the installation and operation of small-scale hydropower plants for the assessed biota were generally small. Construction typically involved clearing a swath of trees approximately 10–30 m wide and 0.5–1.0 km long, covering an area of less than 4 ha. A small footprint, coupled with the siting of the small number of small-scale hydropower plants, resulted in limited environmental impacts on terrestrial biota and instream fishes. Hydropower plants were mainly constructed outside forested gorges, and mostly in areas that were not particularly high in threatened terrestrial species. Further, the plants were mainly constructed in locations where the overlap with fish populations (Atlantic salmon, brown trout and European eel) were likely minimal. Micro plants likely had least individual effect due to smallest footprints, and less impactful infrastructure within the footprint, including the absence of dams. However, land use efficiency increased with plant size class: small plants, the largest of the small-scale classes, had least footprint per unit energy production. Overall, the contribution of small-scale hydropower to national energy policy in Norway is limited due to its relatively low installed capacity and energy production, representing only a small fraction of the total national energy output. Despite a limited contribution to national energy production, small-scale hydropower projects of all size classes (micro, mini and small) can offer benefits to local and regional development by supporting decentralized energy production and local economic growth.

CRediT authorship contribution statement

Richard D. Hedger: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Mahmoud S. Kenawi:** Writing – review & editing, Methodology. **Line E. Sundt-Hansen:** Writing – review & editing, Methodology. **Tor Haakon Bakken:** Writing – review & editing, Methodology. **Brett K. Sandercock:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

Brett Sandercock reports financial support was provided by the Research Council of Norway. The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data availability

Data will be made available on request.

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